



ScubaMind – Mental Health Early Mental Wellbeing Through Passive Insights

ScubaMind is a privacy-preserving mobile app that uses passive smartphone and wearable data to detect early mental wellbeing risk signals. It combines passive sensing, on-device AI, and LLM-based support to provide early awareness with minimal effort while keeping sensitive data local and user-controlled.



Target Problem and Background



Late Awareness

Behavioral changes related to mental wellbeing often remain unnoticed until they become more severe.



High User Burden

Many existing applications rely on frequent self-reporting, manual input, and sustained user effort.



Low Sustainability

Existing tools often struggle to support continuous use, personalization, and meaningful long-term monitoring.



Reactive Support

Most mental health support systems respond after problems emerge rather than enabling early, preventive awareness.





Our Solution and Value Proposition

Our Solution: Passive, Personalized, and Privacy-First Monitoring

1 Passive Data Collection

Steps, sleep, mobility, phone usage, voice, keyboard, and wearable signals.

2 On-Device Analysis

Behavioral features are processed locally using embedded AI models.

3 Personalized Risk Insight

The system generates weekly risk indicators and explanatory feedback.

4 Supportive Interaction

Goals, voice journaling, and LLM assistant chat support enable daily reflection.

Our Value Proposition

Sustainable use: Fits daily routines

Early awareness: Detects changes before crisis

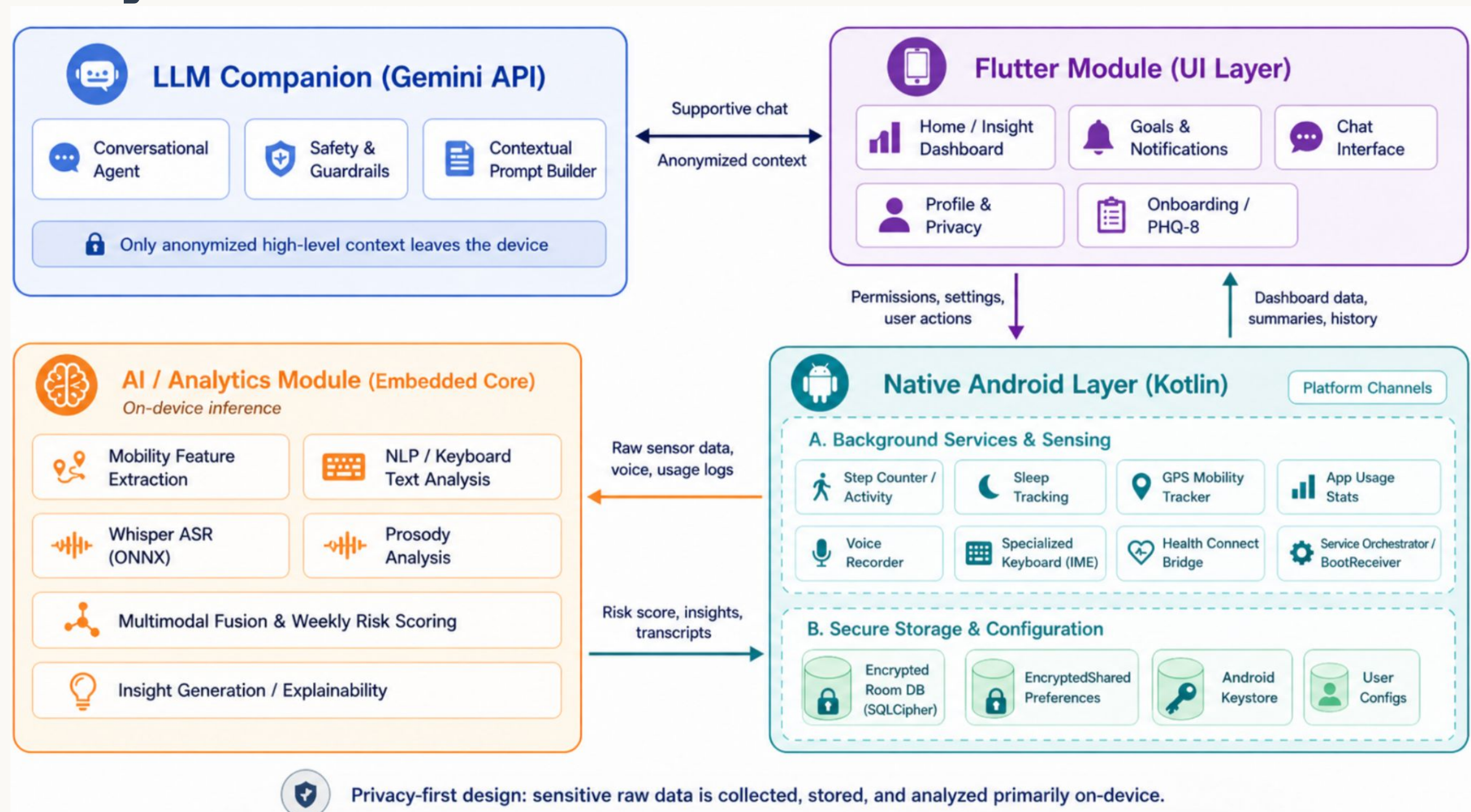
Personalized support: Based on individual patterns

What Makes ScubaMind Different?

- **Passive-first design**
- **Early risk detection**
- **Privacy-aware architecture**



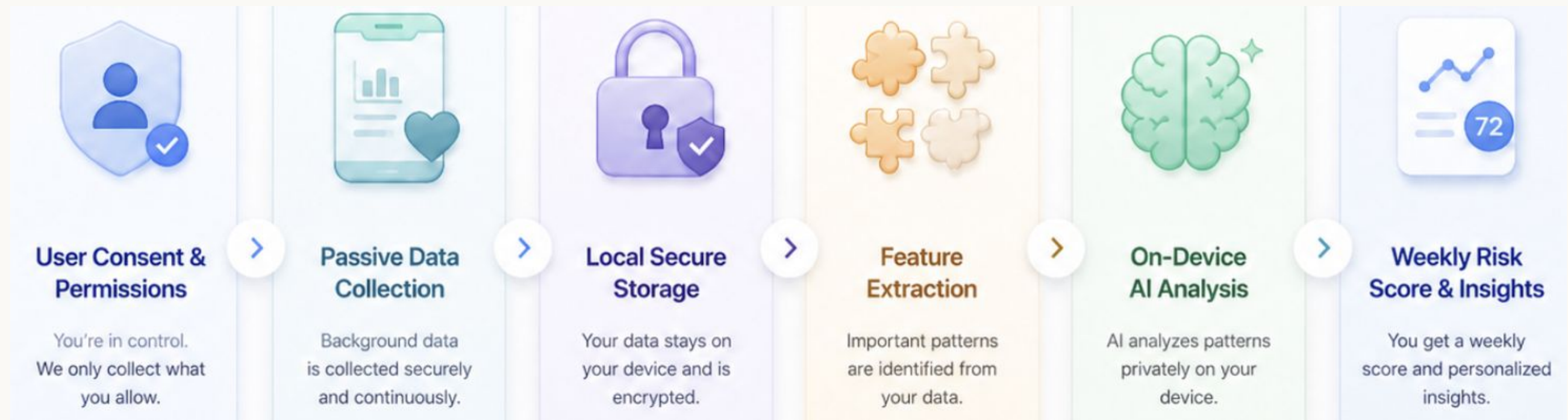
Final System Architecture



[Figure 1 – High-level architecture diagram of the ScubaMind system.]



Data Processing & Pipeline



1. Permission-based sensing

Steps, sleep, GPS mobility, phone usage, voice, keyboard, and wearable signals are collected only with user consent.

2. Unified daily record

Collected data is aggregated into a daily behavioral record for consistent analysis.

3. On-device analysis

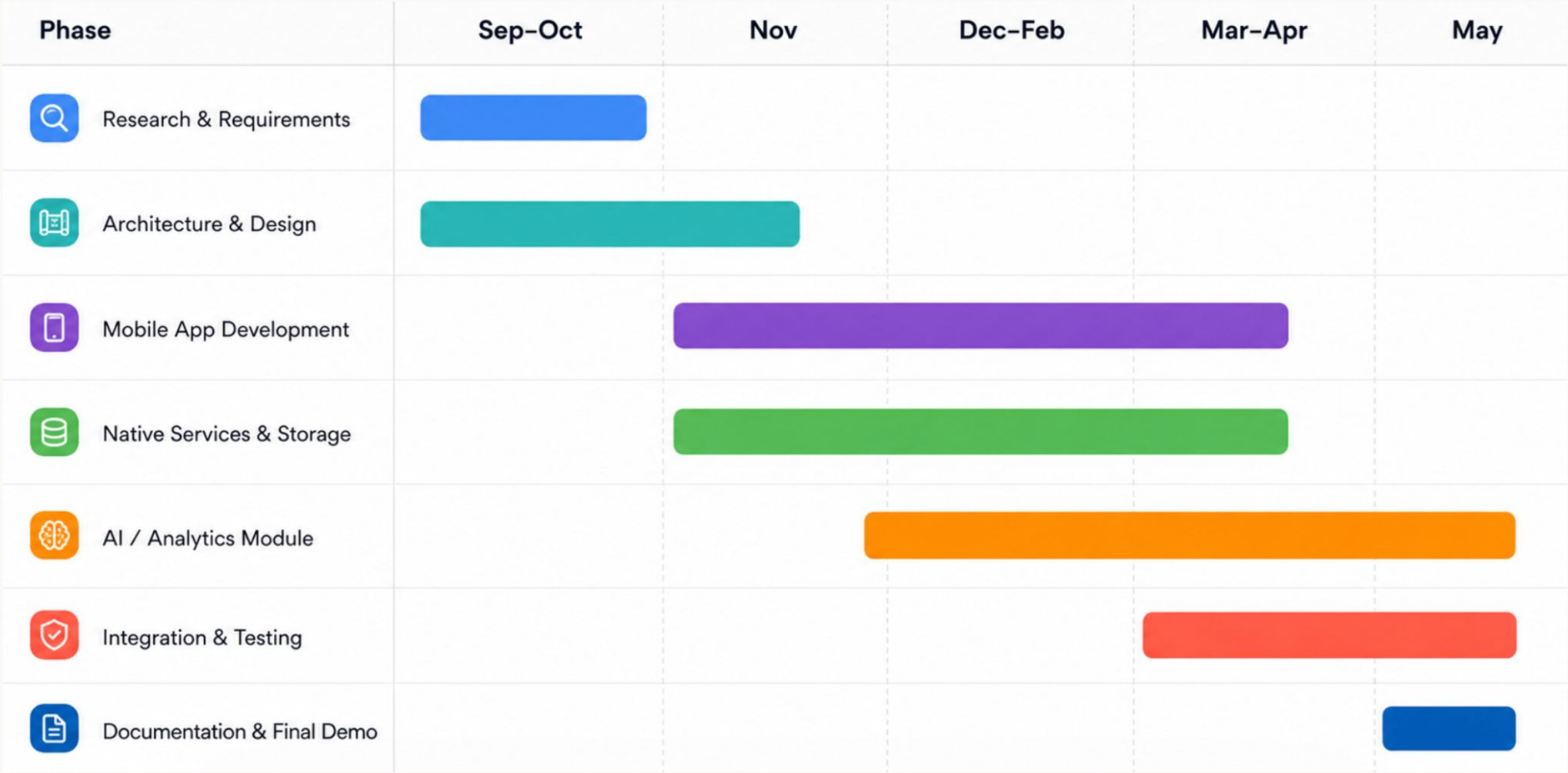
Mobility, sleep, activity, phone usage, text, and voice features are processed locally.

4. Risk insight generation

The system generates weekly risk indicators, explanations, and dashboard feedback.



Project Timeline and Gantt Progress



Final Status

-  Research and design completed
-  Core system implemented
-  Integration and testing completed
-  Final prototype ready for demonstration

[Figure 2 – Project Timeline and Gantt Chart Representation.]

ScubaMind was developed through a staged process including research, architecture design, implementation, integration, testing, and final delivery.



Project Health and Risk Management

Status: Completed and On Track | **Overall Completion:** 100% | **Current Phase:** Final Prototype Completed & CS Fair Preparation

Key Risks & Mitigation

Risk 1 – Sensitive Data Handling

Handled by: Local-first processing, encrypted Room database, user-controlled permissions, and data minimization.

Risk 2 – Reliability of Passive Sensing

Handled by: Foreground services, BootReceiver support, permission checks, and staged testing on Android devices.

Risk 3 – AI Output Responsibility

Handled by: Non-diagnostic wording, PHQ-8 as reference only, explainable insight messages, and LLM safety guardrails.

Ethics & Privacy Status

- Privacy-by-design approach applied
- On-device processing prioritized
- User-controlled permissions implemented
- Sensitive data stored locally and securely
- No critical ethical risks identified

Expert Input & Clinical Guidance

- Regular consultations with professional psychologists.
- Expert feedback supported ethical and user safety.
- Refined non-diagnostic wording and supportive UX.
- Supported privacy, user rights, and responsible use.

Team Contributions

Mobile Application Development (Flutter UI)

Development of the main mobile user interface, including onboarding, home dashboard, goals, profile, privacy settings, and visualization of behavioral insights within the Flutter application.

Responsible: Merve Güleç, Yiğit Koşum, Murathan Işık

Native Android & Passive Sensing Services

Implementation of native Android services for passive data collection and system integration, including activity/step tracking, mobility/location handling, app usage data, background services, and secure local storage support.

Responsible: Merve Güleç, Yiğit Koşum, Murathan Işık

AI, Risk Modeling & Behavioral Analysis

Design and implementation of behavioral analysis pipelines, including risk insight generation, speech/prosody-related analysis, on-device AI support, and multimodal mental wellbeing signal processing.

Responsible: Muhammed Fatih Başal, Metin Çalışkan

System Architecture, Privacy & Integration

Design of the overall system architecture, privacy-first data flow, Flutter–Kotlin integration, local-first processing strategy, and integration of the LLM companion with the final prototype.

Responsible: Team-wide collaboration





Conclusion and Future Work

Conclusion



-  Passive sensing implemented 
-  Weekly risk insights completed 
-  Privacy-first prototype ready 

Future Work



-  Real-world validation 
-  Stronger personalization 
-  Expanded wearable support 
-  Multilingual & iOS support 



ScubaMind demonstrates a privacy-first mobile system for early mental wellbeing awareness.

Thank you — we are ready for the live demo.



Thank You!

ScubaMind – Early Mental Wellbeing Through Passive Insights

We appreciate your attention and interest in our privacy-first approach to early mental wellbeing awareness.

Website

<https://scuba-mental-health.netlify.app/>

GitHub Repository

<https://github.com/yigitkosum/scubamind>

We are ready for the live demo.

